



ya diggg: Sonic Reverse Engineering Spec

This document defines the "Sonic ID" layer for identifying records when the visual DNA (cover/label) is unreadable or missing.

1. The "Sonic ID" Logic (Acoustic Fingerprinting)

When the camera fails to identify a record due to wear, tear, or private press labels, ya diggg activates its **Acoustic Fingerprinting Engine**.

Component	Technical Implementation	Rationale
Capture	High-SNR MEMS Mic (Knowles)	Captures clean audio from speakers or ambient turntable setups
Input	3.5mm Line-In	Allows direct connection to a mixer/turntable for ultra-low noise capture
Processing	OLAF / Chromaprint-lite	Optimized C-library for extracting spectral hashes on ARM Cortex-A53 Github .

2. The "Tiered Sonic Library" Strategy

Since a 100M+ song cloud database (Shazam-scale) cannot fit on a 32GB device, we use a **Tiered Offline Database** [Pcmag](#).

Tier	Size	Description
Tier 1 (Core)	~100MB	Top 50,000 Digger Essentials. Local persistent fingerprints of the most sampled/sought-after records.
Tier 2 (Custom)	~250MB	User Sync Library. Fingerprints pre-cached from the user's Desktop "Wishlist" or "Catalog."
Tier 3 (Cloud)	API Call	Deep Search. Uses the Murata 1DX Wi-Fi module to ping a cloud acoustic database (e.g., AcoustID) for obscure matches.

3. Workflow: "Reverse Identification"

- Visual Failure:** Camera scans unreadable label -> "NO MATCH FOUND."
- Sonic Trigger:** User taps the screen or long-holds the orange button -> "LISTENING..."
- Acoustic Capture:** ya diggg listens for 5-10 seconds of the track.
- Local Match:** The engine hashes the audio and checks the **Tier 1 & 2** local databases
- Reveal:** If a match is found, it reveals the same "Facts-Only" metadata (Personnel, Studio, Sample Lineage).

4. Hardware Updates

To support Sonic ID, the hardware spec is updated with:

- Audio Input:** Dual MEMS Microphones for beamforming noise cancellation in loud record fairs.
- ADC Logic:** The ESS ES9218PC DAC will be configured in **Line-In mode** when the 3.5mm jack is used for input.
- Firmware:** Addition of the **OLAF (Overly Lightweight Acoustic Fingerprinting)** library to the Linux OS layer [Github](#).